

DAY-2

TCP/IP Suite Theory (1 hour): What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access. **Practical (1 hour):** Analyze packet headers to identify the TCP/IP layer using Wireshark. **Task:** Compare OSI and TCP/IP models with real-world examples

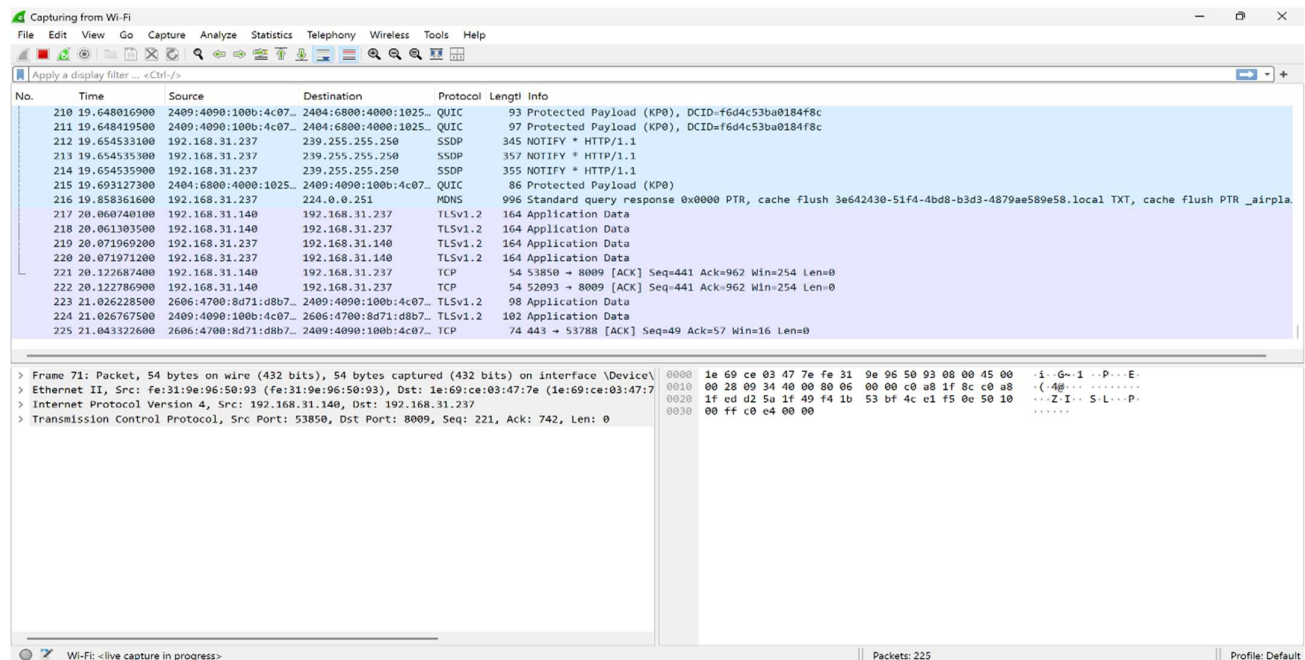
1. Wireshark Packet Capture Analysis

Open Wireshark and start capturing packets. Refresh the Flipkart homepage and observe the captured traffic.

TCP/IP Layer	Example Packet	How Identified
Application Layer	HTTP/HTTPS Packet	Protocol column shows HTTP or TLS/HTTPS traffic
Transport Layer	TCP Segment	Contains Source Port, Destination Port, Sequence Number
Internet Layer	IP Packet	Contains Source IP and Destination IP addresses
Network Access Layer	Ethernet Frame	Contains Source MAC and Destination MAC addresses

Example Observation

- **Application Layer:** HTTPS request sent to Flipkart server.
- **Transport Layer:** TCP packet using port 443.
- **Internet Layer:** IPv4 packet containing source and destination IP addresses.
- **Network Access Layer:** Ethernet frame carrying the packet over the local network.



The screenshot shows the Wireshark interface with a packet capture in progress. The packet list on the left shows several packets, with packet 221 selected. The packet details pane on the right shows the structure of the selected packet, which is a TCP segment. The packet bytes pane at the bottom shows the raw data of the packet.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
210	19.648016900	2409:4090:100b:4c07...	2404:6800:4000:1025...	QUIC	93	Protected Payload (KP0), DCID=f6d4c53ba0184f8c
211	19.648419500	2409:4090:100b:4c07...	2404:6800:4000:1025...	QUIC	97	Protected Payload (KP0), DCID=f6d4c53ba0184f8c
212	19.654533100	192.168.31.237	239.255.255.250	SSDP	345	NOTIFY * HTTP/1.1
213	19.654535300	192.168.31.237	239.255.255.250	SSDP	357	NOTIFY * HTTP/1.1
214	19.654535900	192.168.31.237	239.255.255.250	SSDP	355	NOTIFY * HTTP/1.1
215	19.693127300	2404:6800:4000:1025...	2409:4090:100b:4c07...	QUIC	86	Protected Payload (KP0)
216	19.858361600	192.168.31.237	224.0.0.251	MDNS	996	Standard query response 0x0000 PTR, cache flush 3e642430-51f4-4bd8-b3d3-4879ae589e58.local TXT, cache flush PTR _airpla
217	20.060740100	192.168.31.140	192.168.31.237	TLSv1.2	164	Application Data
218	20.061303500	192.168.31.140	192.168.31.237	TLSv1.2	164	Application Data
219	20.071969200	192.168.31.237	192.168.31.140	TLSv1.2	164	Application Data
220	20.071971200	192.168.31.237	192.168.31.140	TLSv1.2	164	Application Data
221	20.122687400	192.168.31.140	192.168.31.237	TCP	54	53850 → 8009 [ACK] Seq=441 Ack=962 Win=254 Len=0
222	20.122786900	192.168.31.140	192.168.31.237	TCP	54	52093 → 8009 [ACK] Seq=441 Ack=962 Win=254 Len=0
223	21.026228500	2606:4700:8d71:d8b7...	2409:4090:100b:4c07...	TLSv1.2	98	Application Data
224	21.026276700	2409:4090:100b:4c07...	2606:4700:8d71:d8b7...	TLSv1.2	102	Application Data
225	21.043322600	2606:4700:8d71:d8b7...	2409:4090:100b:4c07...	TCP	74	443 → 53788 [ACK] Seq=49 Ack=57 Win=16 Len=0

Packet Details:

- Frame 71: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...
- Ethernet II, Src: fe:31:9e:96:50:93 (fe:31:9e:96:50:93), Dst: 1e:69:ce:03:47:7e (1e:69:ce:03:47:7e)
- Internet Protocol Version 4, Src: 192.168.31.140, Dst: 192.168.31.237
- Transmission Control Protocol, Src Port: 53850, Dst Port: 8009, Seq: 221, Ack: 742, Len: 0

Packet Bytes:

```

0000  1e 69 ce 03 47 7e fe 31  9e 96 50 93 00 00 45 00  ..i.G-1..P...E-
0010  00 28 09 34 40 00 80 06  00 00 c0 a8 1f 8c c0 a8  ..(4@.....
0020  1f ed d2 5a 1f 49 f4 1b  53 bf 4c e1 f5 0e 50 10  ...Z.I...S.L...P-
0030  00 ff c0 e4 00 00
  
```

2. Comparison of OSI and TCP/IP Models

OSI Model Layer	TCP/IP Model Layer	Real-World Example
Application	Application	Sending a WhatsApp message
Presentation	Application	Data encryption in WhatsApp
Session	Application	Maintaining an Instagram login session
Transport	Transport	Reliable delivery of a Zomato order request using TCP
Network	Internet	IP routing to Flipkart servers
Data Link	Network Access	Ethernet or Wi-Fi communication
Physical	Network Access	Electrical signals or wireless radio waves

Example

When ordering food on Zomato:

- Application Layer → Creates the order request.
- Transport Layer → Breaks data into segments using TCP.
- Internet Layer → Adds IP addresses for routing.
- Network Access Layer → Sends data over Wi-Fi or Ethernet.

3. TCP Segment Header Fields

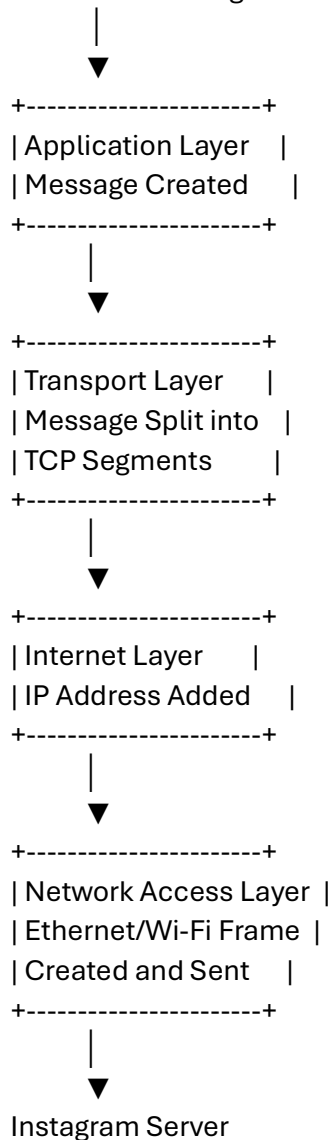
Field	Function
Source Port	Identifies the sending application or service.
Destination Port	Identifies the receiving application or service.
Sequence Number	Keeps track of the order of transmitted data.
Acknowledgment Number	Confirms receipt of data from the sender.
Header Length	Indicates the size of the TCP header.
Flags	Control connection setup and termination (SYN, ACK, FIN, etc.).
Window Size	Specifies how much data can be received before acknowledgment.
Checksum	Detects errors in the TCP segment.
Urgent Pointer	Indicates urgent data that requires immediate processing.

Examples of Common Flags

- SYN → Starts a connection.
- ACK → Acknowledges received data.
- FIN → Ends a connection.
- RST → Resets a connection.

4. Message Flow Through TCP/IP Layers

User Sends Instagram Message



Layer Functions

- Application Layer → Creates and processes the message.
- Transport Layer → Splits data into packets and ensures reliable delivery.
- Internet Layer → Adds source and destination IP addresses.
- Network Access Layer → Transmits data through Wi-Fi or Ethernet.

5. Analysis of One Captured Packet

Selected Packet

HTTPS Packet (TCP Port 443)

TCP/IP Layer

Application Layer

How It Was Identified

- Protocol column displayed **TLS/HTTPS**.
- Destination port was **443**, which is commonly used for secure web traffic.

Real-World Action Represented

This packet could represent:

- Loading the Flipkart homepage.
- Sending a WhatsApp message.
- Booking a ticket on BookMyShow.
- Logging into Instagram.

The packet carries encrypted application data between the user device and the server using HTTPS.